Online Quizzes: The Use of Multiple Attempts and Feedback System

Mark Joseph L. Mendoza
De La Salle Santiago Zobel School
mendozaml@dlszobel.edu.ph

Minie Rose C. Lapinid
De La Salle University-Manila
minie.lapinid@dlsu.edu.ph

ABSTRACT

Timely, specific, and ongoing feedback to correct students' errors and mistakes in an assessment can be overwhelming for teachers, not to mention that in the Philippines, the normal class size is around 40-60 students. In such setup, the feedback system feature in Schoology's online quiz provides instant, and automated feedback to students while answering the assessments. Likewise, the use of this automated feedback system with the multiple attempts feature improved the students' thought process and understanding of Mathematics concepts. This study followed the descriptive mixed methods research design in order to determine the effectiveness of an online quiz with multiple attempts and feedback on students' understanding of sequences and series using the Schoology LMS. A heterogeneous class of forty Grade 10 students participated in this study. These students were exposed to blended learning instruction prior to the administration of the online quizzes. The quantitative phase analyzed the ANOVA Repeated measures results of the online quizzes. The qualitative phase focused on the analysis of the students' solutions in all attempts, and post- assessment reflective journal entries. The study revealed a significant increase across students' online quiz scores which is attributed to the three attempts and automated feedback system. This result was supported by students' reflective journal entries indicating positive attitudes on the use of feedback system such as helping them understand Mathematics concepts, managing time wisely among activities, and becoming independent learners.

CCS CONCEPTS

• **Information systems**; • **Applied computing** → Education; • **E-learning**;

KEYWORDS

learning management system, Schoology, online quiz, multiple attempts, feedback

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1 INTRODUCTION

Timely, specific and ongoing feedback to correct students' mistakes and errors in a quiz can be overwhelming for teachers, not to mention that in the Philippines, a normal class size is around 40 to 60 students. The teacher's repetitive task of checking and writing feedback may be automated using the online quiz features of some learning management systems such as the Schoology, Moodle and Canvas. In Schoology, the online quiz's multiple attempts feature may be used to improve students' thinking and understanding of concepts [30].

The developments in the teaching learning process has unwrapped innovative implementations of teaching techniques and efficient technology-driven learning [19]. Majumdar [24] acknowledge that these changes brought about by the innovations in technology provided opportunities to modify the method of teaching, to attain designed and projected learning objectives, and to visualize a better and improved learning environment suited for 21st century learners.

With these advancements, the absence of physical space and face-to-face instruction between the learners and teachers has led to a variety of assessment delivery methods as a measurement of learning [2]. Assessment is the core of education [1] and a crucial element for effective learning [3]. One of the methods to improve the achievement of the students and the pedagogical practices of the teachers is using formative assessments [10].

1.1 Online Quizzes

The use of formative assessments with feedback system helps improve instructional practices, recognizes possible gaps in the curriculum, and, most especially, contributes to increased student performance [9, 10, 35]. Likewise, formative assessments provide support in monitoring the progress of the students throughout the learning process [36, 37].

Also known as e-assessment, educational assessment deals with the effective use of technology for learning [7]. In a study conducted by Cohen and Sasson [7], they found out that students generally have positive attitudes towards online quizzes using Moodle learning environment as compared to the written tests through the use of a questionnaire [7]. Results of the study indicated that the average grade on both the written tests and online quizzes proved to be significant predictors of their grade in the final exam. Likewise, the research results showed that students also improved their scores significantly and had shorter testing time on the last attempts of the online quiz compared to their first attempts. Further, both learning outcomes and students' attitudes were indicative of the

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instructional design improvement in terms of formative assessment practice [7].

Online quizzes motivate students to complete reading assignments, increase participation in class discussions, and improve performance in exams [6, 9, 20]. The use of online quizzes outside class also give teachers additional time for a more lively, in-depth discussions, higher order thinking and active learning activities in the class [8, 9, 17].

1.2 Distractors

Gierl et al. [12] considered multiple choice type of testing as one of the most widely used and effective form of objective type of test in educational setting because it can directly measure a wide range of knowledge, skills and competencies across disciplines and content areas. This type of test is efficient compared to essays, which are prone to subjective scoring, but may be difficult to construct, specially, the distractors [13]. Distractors are the incorrect options of a multiple choice item "because they are considered 'distracting' to students with partial knowledge due to their plausibility to yield the correct option [12]. According to Gierl et al. [12], there are at least three reasons why distractors are very important part of a multiple choice item: (1) distractors must be plausible but incorrect, thus, must be written by a content expert; (2) distractors form part of the context required to answer the item which affects its quality and learning outcomes; (3) distractor analysis can help test developers and teachers understand why students commit mistakes, thereby guide diagnostic inferences for improved instruction and students' future test performance. These imply that "distractor development is often by content specialists to be the most daunting and challenging component of writing a multiple choice item" [12].

As mentioned by Gierl et al. [12], two general strategies of distractor development have been consistently described and advocated: (1) create distractors using common misconceptions or errors in thinking, reasoning and problem solving, and (2) create distractors similar in content and structure relative to the correct option [12]. In the case of numeric options, for example, once the correct answer is calculated, factors can be removed or inverted and the answer recalculated [12]. Further, distractors can also be from the same category as the correct option, such as the same concept, topic, or idea [12]. Gierl et al. [12] reiterated that all of these fall under content similarity. Structural similarities include distractors that have the same characteristics as the correct option in terms of length, complexity, formatting and grammar.

1.3 Feedback

Since learning is sequential and continuous, the power of feedback plays an important role in achieving better learning outcomes to establish powerful impacts on learning [15]. In education, feedback is an essential element to enrich the teaching-learning process [33] and a crucial tool that can be used by teachers in improving the knowledge and skill acquisition of the learners [25]. Thus, as stated by Voerman et al. [33], feedback has a powerful but variable influence on learning.

According to Hyland [18], feedback serves a variety of purposes including evaluation of students' achievements, development of students' skills and understanding, and elevation of students' motivation and confidence. As mentioned in the study of Hatziapostolou and Paraskakis [16], Butler and Winne stated that feedback is a key element in regulating self-learning because it enables learners to monitor their development and adjust their strategies as they progress towards their learning goals. Further, feedback helps teachers to better set targets for their students, creates independent student learners and, in the process, raises students' performance levels [21].

As cited in the research of Hattie and Timperley [15], Kulhavy clarifies that feedback is not necessarily a reinforcement because it may be accepted, modified or rejected by the learner depending on how they interpret, understand and evaluate the feedback. Hattie and Timperley [15] reiterated that if feedback is not properly communicated by the teacher and incorrectly translated by the receiver, it may not have the power to initiate further action. Thus, specific and easy to understand feedback becomes a vital element in providing responses to further clarify and verify these misconceptions [15].

Pauli [28] found the most common form of feedback given by teachers are usually not specific such as "Well done!" and "Very good!". As cited by van der Hulst, van Boxel, & Meeder [32], Nicol pointed out that an effective teacher feedback, in any form, should be understandable, selective, specific, timely, contextualized, non-judgmental, balanced, forward looking, and transferable. Likewise, feedback is most effective if the learning goals are specific and challenging [15].

In education, the untimely feedback poses a problem [35]. Since most teachers have far too many students, it is common that a vital feedback is given in days, weeks or even in months after the assessment or performance [11]. Due to the delayed feedback, students repeat the same mistake again [11]. Thus, educators should find ways to provide timely feedback so students could use it effectively during the learning process [35].

In addition, feedback must be ongoing [35]. A useful feedback must not only be received by the student but should also be immediately used to improve formative understanding of concepts [5]. Thus, it is not too late to give feedback as the student still has the opportunity to learn and ultimately perform better in the summative assessment [35]. In addition, feedback creates an avenue for self-evaluation which allows the students to self-edit their own work [29].

Fisher and Frey (2012) provided some guidelines for teachers to save time in giving effective feedback so teachers could create more time for learning [11]. Among these tips are to identify patterns in student errors and to guide learners towards increased understanding [11]. Instruction and intervention in feedback on specific areas, rather than re-teaching the entire lesson enables teachers to be more precise in addressing the error [11].

The use of digital technology in providing feedback has potentially raised the feedback quality and saved teachers' marking time [32, 35]. As cited in the research done by Yusof et al. [3], Guardado and Shi mentioned that electronic feedback reduces the problems in terms of logistics at the same time maintains some of the best features of traditional written feedback [3]. Further, van der Hulst, van Boxel, & Meeder [32] pointed out that digital-based feedback (1) is clear and comprehensible since teachers' handwriting are sometimes illegible, (2) could be accomplished faster, (3) reused and reposted by teachers, and (4) timely as they can be delivered faster to the receiver.

In their study, van der Hulst, van Boxel and Meeder [32] pointed out the significance of maximizing digital technology in providing feedback. They identified different advantages of using digital technology in providing feedback which include the (1) formulation of readable, effective, and comprehensible comment or guide on a certain assessment; (2) ability to provide direct and specific annotations; (3) potential to serve as a guide for learners as they go through schoolwork in which objectives are aligned with specific criteria; (4) quick delivery of instant and immediate feedback in digital format which creates an opportunity for the students to frequently go back to the feedback for concept and skills check-up; and (5) opportunity for teachers to re-use and re-post feedback on any work done by the learners especially if the activity covers the same learning goals, thereby giving the teacher more other time in preparing other classroom activities [32].

The use of online and blended learning has developed significantly to address the needs of the 21st century learners [2]. Likewise, the advent of Learning Management System (LMS) has provided instructors with ways on how to electronically create and deliver contents and feedback, to monitor student involvement and engagement, and to assess student performance and achievement [23].

Throughout the materialization of different education technology tools, Hatziapostolou and Paraskakis [16] stated that the effectiveness of feedback could be augmented when it is communicated via the student's desktop or personal learning space which is referred to as the LMS, a learning environment where all learning materials and resources are stored. Further, feedback as well as comments are two of the best features of an LMS which are crucial in the online teaching learning process [14, 22, 23].

1.4 Conceptual Framework

The feedback system utilized in this study considered the key elements of effective technology: clarity and formulation of feedback, annotations, alignment with assessment criteria, personalized feedback, timeliness of feedback and time saving for teachers, of which as argued, can only be possible with the use of the digital technology [32]. As explained by Wiggins (201), multiple attempts provided opportunity for the students to correct their mistake which is a crucial element of an effective feedback [35].

As discussed by Gierl et al. [12], the two general strategies of distractor development which are (1) using common misconceptions or errors in thinking, reasoning and problem solving, and (2) creating distractors similar in content and structure relative to the correct option ensure that distractors are plausible, thereby requiring an effective feedback to address the error and misconception. On the other hand, a positive reinforcement should appropriately be given to the student for the correct option in each multiple choice type item. All these are aimed to produce in depth, higher other thinking, active learning, and improved performance among students [12].

The study aimed to determine the effectiveness of online quiz with multiple attempts and feedback on students' understanding of sequences and series using the Schoology LMS. Specifically, it sought to answer the following questions:

What are the students' understanding of sequences and series based on their scores in the online quizzes with feedback system, in the three attempts?

How do students perceive the use of feedback system in online quizzes?

2 METHODS

2.1 Research Design

This study employed a descriptive mixed methods research design in investigating the use of online quiz with multiple attempts and feedback system in improving students' understanding of sequences and series. The quantitative data consist of test scores and the qualitative data consist of students' solutions and reflective journal entries.

2.2 Research Participants

Forty Grade 10 students in an intact heterogeneous class from a private Catholic school in the Metropolitan area in the Philippines participated in the study. Students had familiarity in using Schoology since the LMS had been used in the lessons prior to sequences and series. The heterogeneous class was specifically chosen because of its diversity and class schedule which fits the first researcher.

2.3 Research Instruments

The research instruments used were the online quizzes on sequences and series, and online reflective journal. These instruments were face and content validated and evaluated by three mathematics teachers, mathematics coordinator, and education technology coordinator. All validators and the first researcher had units in master's degree and were also grantees of certifications of excellence in educational technology awarded by Microsoft Corporation and Google.

2.3.1 Online Quiz and Feedback System. The researcher used Schoology's online quiz feature as one of the assessment tools in the study. This served as the formative assessment that covers arithmetic sequences and series, geometric sequences and series, harmonic and Fibonacci sequence, and applications of sequences and series. Four online quizzes were prepared and created allowing students three attempts. In the second and third attempts, the students were required to re-answer all the items in the previous attempt in order to correct the missed or incorrect items and to double check the mastery of concepts and skills for the correct items. In each attempt, Schoology rendered the total score and recorded the highest among the three attempts. Test items were aligned to target competencies on sequences and series. Just like the achievement test, the online quizzes were of multiple choice type with four options. The 1st, 2nd, 3rd and 4th online quizzes had a total of 14, 14, 10 and 10 points, respectively, and were administered in 30, 30, 25, and 25 minutes, respectively.

Schoology online quiz does not only allow the administration of online teacher-made assessments but also the formulation of specific online feedback per item [30]. Through this, the student participants were guided on which items were answered correctly

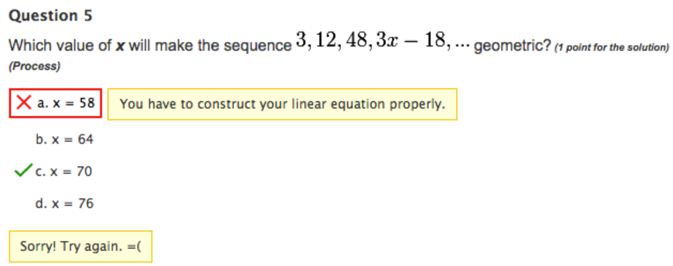

Figure 1: An example of feedback for an incorrectly answered item

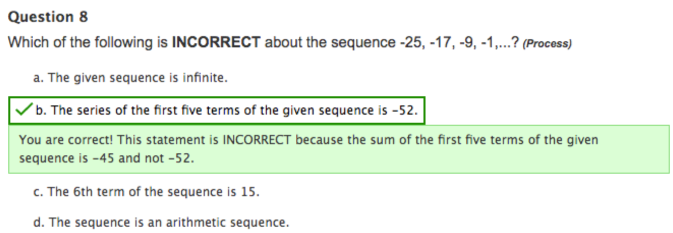

Figure 2: An example of feedback for a correctly answered item

and incorrectly. The feedback system in the online quiz was also used to further facilitate assistance and instruction to the students while taking the online assessment. The teacher phrased the feedback in each option based on its plausibility qualifying the option a good distractor.

Figure 1 shows the feedback for the fifth question of online quiz 2 on geometric sequences and series. Students were prompted with the error in their computation if the chosen answer was option a. As shown in the figure, the student might have constructed the linear equation incorrectly leading him/her to an incorrect answer. Since the correct answer was shown in the feedback, students were required to submit their solutions to each attempt to track the progress and changes made in the solutions in accordance to the provided feedback.

Figure 2 shows an example of a positive reinforcement feedback provided for correct answers. For the correct items, the usual feedback were "*Good Job!*", "*Great Work!*", or "*Congratulations!*". For the incorrect items, students were given comments on how to correct the committed errors. Some of these feedback were "*Double check the value of your common difference.*", "*Please review combining like terms.*", or "*Check the value of x.*"

2.3.2 Online Reflection Journals. Students were asked to fill out online reflective journals via Google Forms to document their learning experiences while answering the online quiz with multiple attempts and their thoughts about the feedback system. The questions asked were: (1) How did you find the feedback in the online quizzes in Schoology? Was it helpful identifying the skills that you need to improve on? (2) How did you use the feedback in the online quiz in making sure that you will get the correct answer when you re-answer an item? and (3) Evaluate the overall effectiveness of the feedback system in Schoology Online Quiz. Explain why you chose that overall evaluation.

2.4 Data Gathering Procedure.

After getting permission from the principal to conduct the study, parents' consent of the students was secured. In each lesson, the first researcher who was also the teacher of the class, discussed the concepts and skills using blended learning instruction, a combination of the traditional face-to-face lectures with online classes. One hour was allotted for each face-to-face meeting with an integration of an online self-paced instruction. The students were asked to work on online activities posted in Schoology. After each lesson, an online quiz with multiple attempts was administered. There were 4 online quizzes, one online quiz for each lesson. Students were asked to accomplish the online reflective journals using Google Forms for an hour after the implementation of all four lessons.

2.5 Data Analysis.

Students' scores in the three attempts of each quiz were analyzed using ANOVA repeated measures, after all conditions warranting the use of this parametric test had been checked and satisfied. Students' solutions were examined to determine the authenticity of their understanding and learning. Likewise, since the correct answers were shown in the feedback after an attempt, the solutions were used to rule out the possibility of just copying the correct answer in the succeeding attempts.

Lastly, students' online reflection journal entries on the use of the feedback system were analyzed using thematic analysis [4]. Braun and Clarke's [4] six-phase framework of thematic analysis was followed: (1) familiarize the data, (2) generate initial codes, (3) search for themes, (4) review themes, (5) define themes, (6) write-up. Verbatim written responses were carefully read and re-read going from one student to the next. Useful notes were jot down for early impressions. Data were then organized in a meaningful and systematic manner through codes based on the researcher's

Table 1: Summary of Students' Perception on the use of feedback system in online quizzes

Common Themes	Frequency	Percentage
Guide in answering the test item	25	62.5%
Helpful in understanding the test item	17	42.5%
Still needs assistance from or clarification from a teacher	2	5%
No response	1	2.5%

7. What is the sum of the first ten terms of the sequence −25, −17, −9, −1, ...? *(1 point for the solution) (Process)*

OPTIONS	FEEDBACK
a. *−485*	Are you sure that your common difference is correct?
b. 610	Double check your computations. You missed something important when you apply the basic operations.
c. *110*	Your answer is correct!
d. *235*	Ooops! Please check your formula!

Figure 3: Feedback System of Item #7 of Online Quiz 1

perspective aimed at answering the research question. The codes were collated into initial emerging themes. Some codes fit into one theme, others to two or more themes. The frequency count of responses per student falling under each theme was determined. These themes were reviewed in the context of the entire data set to make sure that they make sense and data based. In each theme, data were gathered and exemplary responses were chosen to best represent the theme.

3 RESULTS AND DISCUSSION

The analysis of the data was done in two phases: the qualitative phase and the quantitative phase. The qualitative phase of the study used thematic analysis to analyze the online reflection journal entries of the students on the use of the feedback system. On the other hand, the quantitative phase used ANOVA repeated measures to analyze the students' scores in each of the three attempts of the four online quizzes.

3.1 Qualitative Phase: Student's Perception on the use of feedback system in Online Quizzes

Table 1 presents the summary of the student's perception on the integration of the feedback system of Schoology online quiz.

As shown in the table, most of the students found the feedback system to be a helpful tool in understanding the lesson. As can be gleaned from Table 1, majority (62.5%) considered the feedback as a useful guide in answering the items. They were prompted to re-compute and recall taught concepts. The excerpts below were taken directly from the students' journal reflections to illustrate this:

Student #8: "*Overall, the feedback system allowed me to clearly see which aspect of the lesson and what set of skills I need to improve on. It allowed me to correct my mistakes and learn from them.*"

Student #14: "*The students get to have their mistakes corrected even if the teacher wasn't there.*"

Student #21: "*It's very useful since it really show and explain where I went wrong and how I could correct it.*"

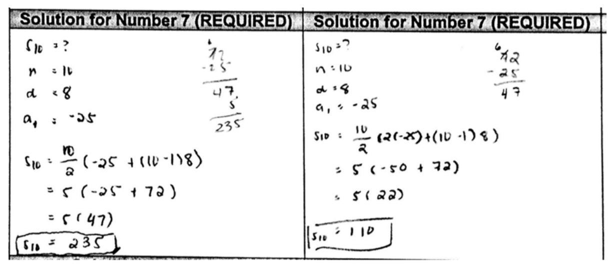

Figure 4: Solution of Student #6 in the first 2 attempts.

Further, seventeen students appreciated the feedback system because it helped them understand the skills and concepts more which are needed in answering the given problem. For example, Student #16 said: "*It helps me a lot in understanding the problem better.*" To further support the claim in students' reflective journal and their attempts, Figure 3 presents item #7 of the first online quiz.

Figure 4 shows the solutions of Student #6 for item #7 of online quiz 1.

As shown in the figure, the student answered option D which is incorrect with a feedback on the use of the correct formula. The solution indicated that he missed the formula of multiplying the first term, -25, by 2. In the 2nd attempt, he considered the given feedback and was able to answer the item correctly.

Figure 5 shows the question, options, and feedback for Item #5 of online quiz #2.

Figure 6 shows how the feedback helped the student to correct his mistake for item #5 of online quiz #2 as shown in Figure 5. Student #1's first attempt resulted to 58. This was due to his error in the transposition of the term -18. The student did not solve the linear equation as he did in his first attempt. In the second attempt, the student changed his strategy in solving the correct answer. Instead of following the feedback, he applied the concept of substitution in solving the item. By elimination method after the feedback in the first attempt, he tried the next option which was 64 (as shown in the erasure), and finally got the correct answer using the value 70.

Student #1's solution imply that feedback can be suggestive and not prescriptive. Feedback can serve as a guide, but the student has

5. Which value of $\boldsymbol{x}$ will make the sequence $3, 12, 48, 3x - 18, \ldots$ geometric? *(1 point for the solution) (Process)*

OPTIONS	FEEDBACK
a. $x = 58$	Do not forget to change the signs when you transpose terms!
b. $x = 64$	Ooops! Please try again. You missed out a term in the process.
c. $x = 70$	Congratulations! You are correct!
d. $x = 76$	Ooops! Recall the process on how to solve linear equations!

Figure 5: Feedback system of Item #5 of Online Quiz 2

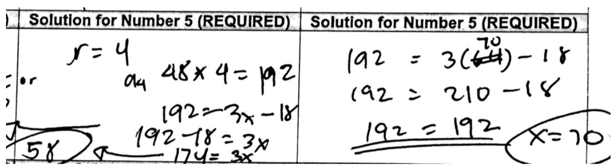

Figure 6: Student #1's solution for Item #5 of Online Quiz #2

9. What is the ninth term of the sequence $1, -3, 9, -27, 81, -243, \ldots$? *(1 point for the solution) (Process)*

OPTIONS	FEEDBACK
a. $a_9 = -19683$	Ooops! Double check your computations. We're looking for the 9th term only!
b. $a_9 = -6561$	Analyze the pattern of the signs in the sequence. Is your common ratio correct?
c. $a_9 = 6561$	Your answer is correct.
d. $a_9 = 19683$	Sorry! Try again! Double check your formula!

Figure 7: Item #9 of Online Quiz 2

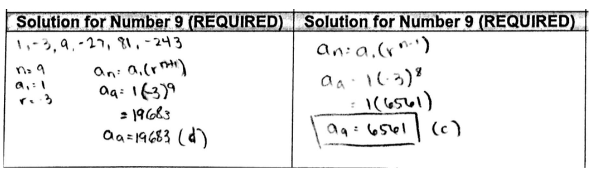

Figure 8: Solution of Student #12 for Item #9 of Online Quiz 2

a considerable amount of freedom to do his own way to come up with the correct answer, which may be different from the teacher's.

Figure 7 shows the question, options, and feedback for Item #9 of online quiz #2.

As can be gleaned in Figure 8, student #12 committed a mistake in the application of the formula of the geometric sequence. In her first attempt, she used $a_n = a_1(r^{n+1})$ instead of the correct formula which is $a_n = a_1(r^{n-1})$. As shown in the figure, this was corrected in her second attempt.

The teacher painstakingly took careful consideration on how students could possibly answer the questions. Thus, the choices were all plausible as these were validated by the content experts. Consequently, the teacher's role as moderator and facilitator of learning could be seen in the process of designing, maximizing, and integrating the different features of LMS in a blended learning environment [26].

Students' scores and solutions all suggest improvements in their understanding of the concepts and skills. Their solutions in the different attempts were testaments on how they were able to apply the feedback in deciphering their mistakes and correcting them. This finding confirmed previous studies [32, 34, 38] that feedback creates an avenue for meaningful and significant interactions between students and teachers. If feedback provides opportunities for learners to act on it, then a direct, timely, understandable, and readable feedback is mostly effective [32].

Nonetheless, there were two respondents who still needed assistance or clarification. Furthermore, Fisher and Frey [11] cautioned that errors could be due to students' lack of knowledge. Despite being alerted and given feedback, the student simply did not know what to do to fix the problem due to the lack of skill or conceptual understanding [11]. This further confirms that technology and its features are not replacements for schools or teachers [27]. In this case, there was a need for the teacher to reteach the lesson to these two students.

3.2 Quantitative Phase: Student's Scores in Online Quizzes

Table 2 shows the students' mean scores and standard deviations in the four online quizzes across the three attempts. The 1st, 2nd, 3rd and 4th online quizzes had a total of 14, 14, 10 and 10 points, respectively. All of the four online quizzes show an increasing pattern in the mean scores across the three attempts. The lowest mean score is 8.40 points for online quiz 1 which is 60% of the total score. Students in this attempt barely passed since the school required a passing rate of 60%. Through the use of the online feedback in Schoology, students were guided on how to correct their mistakes. Students' mean score increased to 10.175 in the second attempt and eventually to 11.650 in the third attempt. Students' scores were more spread and varied in their first attempt and eventually became more consistent in the 2nd and the 3rd attempts, except for the first online quiz. The increase in the mean score in the 1st online quiz caused a wider spread in the scores from the first to the second

Table 2: Mean and Standard Deviation of Online Quiz Scores across the Three Attempts

Online Quiz	Topic	Attempt	Mean	Standard Deviation
1	Arithmetic Sequences and Series	1	8.4000	1.8920
		2	10.175	1.9333
		3	11.650	0.8336
2	Geometric Sequences and Series	1	9.275	3.1049
		2	11.525	2.1601
		3	13.250	1.4806
3	Harmonic and Fibonacci Sequence	1	9.100	1.3737
		2	9.375	0.9251
		3	9.900	0.3789
4	Applications of Sequences and Series	1	9.225	1.2297
		2	10.000	0.0000
		3	10.000	0.0000

Table 3: ANOVA Repeated Measure Results of Students' Online Quiz Scores

Online Quiz	Source	Type III Sum of Squares	df	Mean Square	F	Sig.
1	Attempts – Greenhouse Geisser	211.850	1.722	123.011	72.380	.000*
	Error Attempts – Greenhouse Geisser	114.150	67.166	1.700		
2	Attempts – Greenhouse Geisser	317.850	1.512	210.283	76.764	.000*
	Error Attempts – Greenhouse Geisser	161.483	58.950	2.739		
3	Attempts – Greenhouse Geisser	13.217	1.141	11.582	18.333	.000*
	Error Attempts – Greenhouse Geisser	28.117	44.504	.632		
4	Attempts – Greenhouse Geisser	16.017	1.000	16.017	15.888	.000*
	Error Attempts – Greenhouse Geisser	39.317	39.000	1.008		

*Significant at $\alpha = 0.05$

attempt. A plausible explanation could be that the students were still adjusting and getting used to how they could properly correct their mistakes through the use of the feedback system. Remarkably, all students were able to get the perfect score in the 4th quiz's 2nd and 3rd attempts.

Table 3 presents the results of the ANOVA repeated measures tests of these mean scores. In the first online quiz, $F(1.722, 67.166) = 72.380$, $p < 0.05$, there is an evidence to reject the null hypothesis and conclude that there is a significant increase in the online quiz scores of the students across all the three attempts. The same can be deduced from the ANOVA test results of the other 3 online quiz scores.

4 CONCLUSION

The recorded significant increase in all four online quiz scores across the three attempts is an indication that the feedback system associated to each item generally helped students to correct their mistakes, to eventually understand the concepts and improve their computational skills.

This result is further supported by the reflective journals where the students acknowledged the value of the feedback in the online quizzes. Most students found the feedback as an effective guide on how to correct their mistakes. Students were able to follow the instruction in the feedback without the physical presence of the teacher. This implies that the teacher had saved a considerable amount of time from going over students' mistakes and writing his feedback. The feedback were typewritten and can't be misinterpreted due to illegible handwriting. Students viewed the feedback very helpful in their understanding of the test item. Not being able to comprehend the given test item was very probable, especially when students were aware that they were taking the test and were under pressure to perform. Providing the timely feedback and the multiple attempts helped them to take a second look at the test item and approached it in a different way as their previous attempts.

In order to discourage students from guessing and to seize the opportunity to learn from the feedback, students were required to show complete solutions in answering the online quiz in each attempt. This allowed the teacher to follow through students' progress in understanding the required concepts and skills and to verify the validity of the feedback in students' perspective.

On the other hand, there were students who did not follow the given feedback and had a different way in answering the test item. Nonetheless, students were not jeopardized if they did not follow the feedback's advice. In this sense, students had some degree of autonomy in answering the test. Since a problem can be solved in various ways and students had their own preferences in solving mathematical problems [31], the feedback should not be too restrictive and imposing. In order to not inhibit students from their own thinking and decision making, the teacher may make it clear to the class that feedback is only suggestive, and neither prescriptive nor mandatory.

Despite the positive outcome of the study, two students did not find the feedback helpful in correcting their mistakes or errors. Fisher and Frey [11] suggested that teachers should observe patterns in students' errors to target instruction or intervention on specific areas that students need rather than reteach the entire concept. However, based on their views of the feedback and poor performance in the online quizzes, these two students were given more attention through additional consultation meetings with the teacher for remediation.

5 RECOMMENDATION

Despite careful planning in the conceptualization and implementation, this study is not without fault. Based on the results and conclusions, the researchers suggest conducting further studies:

- consider giving feedback based on the four broad categories of errors enumerated by Fisher & Frey [11]: factual, procedural, transformation and misconception errors;
- examining how timed and untimed online quizzes should be integrated with the multiple attempts and the feedback system features of an LMS; and
- examining what scoring scheme could best fit online quizzes with multiple attempts and feedback system: Should the teacher record (1) the highest score among the three attempts, (2) the score of the last attempt or (3) the average of all these attempts?

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